

FINAL PRESENTATION · THREE-STUDY PROGRAM

Vocode: Efficient Neural Vcoders

A twelve-model study of **efficient speech synthesis** across audio quality, inference speed, and model footprint.

RESEARCH STRUCTURE

01

Budget-controlled reproduction

Project-trained baselines under explicit compute budgets

02

Architecture-aware optimization

Supported interventions applied to frozen checkpoints

03

Cross-model comparative study

Quality, complexity, inference, reproduction, and optimization trade-offs

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AGENDA

Research Sequence



STUDY
PROGRESSION

STUDY 1
Baseline reproduction

→ STUDY 2
Checkpoint optimization

→ STUDY 3
Cross-model synthesis

LJSpeech 1.1 Dataset

DATASET USED

LJSpeech 1.1

Single-speaker English read speech drawn from seven public-domain nonfiction books. The recordings were produced by LibriVox and aligned with text by Ito and Johnson.

Public-domain corpus · released in 2017

Data characteristics

13,100

UTTERANCES

23:55:17

TOTAL DURATION

22.05 kHz

MONO · 16-BIT PCM WAV

1.11–10.10 s

CLIP DURATION · MEAN 6.57 S

Text annotation: utterance identifier, verbatim transcript, and normalized transcript with manual audio-text quality checking.

Train, validation, and test split

13,100 UTTERANCES IN TOTAL

TRAINING

12,475 95.2%

VALIDATION

100 0.8%

TEST

525 4.0%

SOURCE Ito, K. and Johnson, L. (2017), The LJ Speech Dataset; project split: 12,475 / 100 / 525 utterances.

Data Preprocessing

SHARED WAVEFORM PATH

From corpus record to model input

- 01

Validate and index

metadata.csv · id | raw text | normalized text

Check corpus paths and nonempty fields; resolve WAV records; sort identifiers.
- 02

Decode waveform

int16 PCM → mono float32 [-1, 1]

Load lazily; average channels if present; divide sample values by 32,768.
- 03

Apply recipe transforms

gain → peak 0.95 → resample → crop / pad

Each operation is conditional; resampling precedes segment extraction.
- 04

Separate training from evaluation

TRAINING

Shuffle from seed 1234; random fixed-length crop or right-pad; optional gain.

VALIDATION / TEST

Fixed order; complete utterance; no random crop or gain.
- 05

Collate and condition

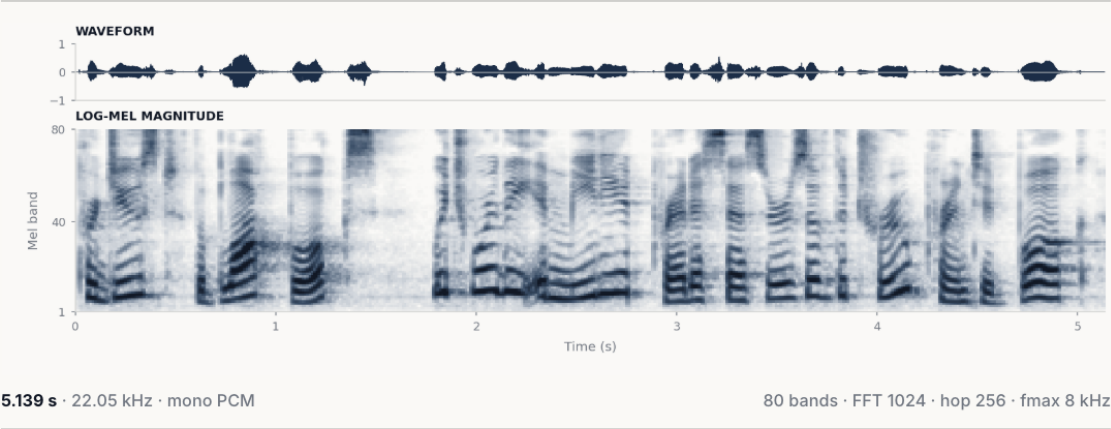
right-pad batch · retain true waveform lengths

Models compute their own mel or conditioning features; no universal mel is imposed.

SIGNAL REPRESENTATION

Recorded waveform and reference mel

LJ001-0004



Reference-recipe controls

WAVEFORM RATE	16 / 22.05 / 24 kHz, selected by architecture.
MEL CHANNELS	80 / 100, selected by architecture.
VARIABLE FIELDS	Frequency ceiling; STFT centering; padding; mel scale; basis normalization.

Reproducibility controls

TRAINING	Seed 1234
EVALUATION	Seeds 42 / 43 / 44; one retained checkpoint; repeated inference only.

STUDY 1 · RESEARCH PROBLEM

Budget-Controlled Training Reproduction

RESEARCH QUESTION

Under explicit project budgets, how do twelve vocoder configurations trained from random initialization trade synthesis quality, real-time factor, and model complexity?

WHY PAPER SCORES ARE NOT DIRECTLY COMPARABLE

Reported metrics are protocol-dependent

- Checkpoint selection
- Training exposure
- Sample rate and conditioning
- Evaluator and hardware

Study 1 aligns these factors within one registered evaluation protocol.

OPERATIONAL TEST

Registered training and evaluation protocol

01

Model cohort

12 configurations · 5 synthesis families

Published architectures plus one literature-derived controlled adaptation.

02

Training

Random initialization · seed 1234

Architecture-specific recipes under registered compute budgets.

03

Inference protocol

525 utterances · seeds 42 / 43 / 44

Repeated inference from one retained checkpoint per model.

04

Headline metrics

PESQ / STOI · B200 RTF

Parameter count · serialized size

Supplemented by spectral, pitch, voicing, and MAC diagnostics.

CONTRIBUTION

Twelve project-trained baselines spanning five synthesis families, reused unchanged in Studies 2 and 3.

CLAIM BOUNDARY

Project-budget evidence only; no author-scale convergence, equal-compute, or architecture-invention claim.

STUDY 2 · RESEARCH PROBLEM

Architecture-Aware Optimization and Deployment

RESEARCH QUESTION

Which architecture-appropriate interventions reduce latency, serialized size, or arithmetic cost while preserving synthesis quality across the twelve frozen Study 1 checkpoints?

WHY TRANSFORMATION LABELS ARE INSUFFICIENT

A transformed model must execute on its target backend

- Executable operator coverage
- Deployable artifact identity
- Same-stack baseline
- Quality and latency measured together

Study 2 resolves benefit at the model, operator, and backend level.

OPERATIONAL TEST

Applicability-aware intervention protocol

01

Frozen baselines

12 Study 1 checkpoints · SHA-256 bound

No baseline retraining; one retained state per configuration.

02

Applicability gate

Architecture · operators · backend

Supported, unsupported, and measured no-op cells remain explicit.

03

Intervention matrix

Quantization · pruning · compilation

Reduced precision, ONNX Runtime, and ODE step reduction where applicable.

04

Evaluation

525 utterances · seeds 42 / 43 / 44

Quality, RTF, latency, size, coverage, and complexity measured per cell.

CONTRIBUTION

A controlled comparison of how supported optimizations change audio quality, inference speed, and model size.

CLAIM BOUNDARY

Results apply only to the tested model, checkpoint, and backend; no optimization is claimed to work universally.

STUDY 3 · RESEARCH PROBLEM

Controlled Cross-Model Comparison

RESEARCH QUESTION

Which of the twelve project-trained vocoder designs offers the best balance of synthesis quality, inference speed, and model complexity under one evaluation protocol?

WHY ONE METRIC IS INSUFFICIENT

The best model depends on the deployment constraint

- Synthesis quality
- GPU and CPU latency
- Parameter count and model size
- Training and optimization context

Study 3 compares trade-offs instead of declaring one universal winner.

OPERATIONAL TEST

Model-to-model comparison under one protocol

01

Evidence cohort

12 project-trained vocoder designs

Author checkpoints do not enter the measured ranking.

02

Common protocol

525 utterances · seeds 42 / 43 / 44

Fixed LJSpeech split and isolated timing for every admitted row.

03

Comparison axes

Quality · RTF · parameters · model size

Training budget and optimization results remain visible for each model.

04

Decision rule

Pareto frontier · deployment constraint

Select by fidelity, latency, memory, or CPU requirements.

CONTRIBUTION

One controlled comparison showing which models fit different quality, speed, and size requirements.

CLAIM BOUNDARY

Conclusions apply to these twelve project-trained models on LJSpeech and the measured hardware.

ARCHITECTURE LANDSCAPE · 12 CONFIGURATIONS

Five Synthesis Families

Twelve project-trained configurations grouped by the operator that reconstructs the waveform.

FAMILY		PROJECT CONFIGURATIONS	SYNTHESIS FORMULATION		
01	Time-domain GAN LEARNED TEMPORAL UPSAMPLING	HiFi-GAN V1 / V2 / V3 · MelGAN · BigVGAN	$\hat{x} = G_{\theta}(m)$		
02	Frame-level spectral COMPLEX SPECTRUM AND ISTFT	Vocos · APNet2 · FreeV · RNDVoC	$\hat{x} = \text{iSTFT}\left(\hat{A} \odot e^{j\hat{\phi}}\right)$		
03	Attention-augmented spectral FRAME ATTENTION AND SPECTRAL HEAD	VocosFormer	$h = \text{ConvNeXt}_{\theta}(\text{Attn}_{\theta}(m))$ $\hat{x} = \text{iSTFT}(S_{\theta}(h))$		
04	Rectified flow MULTI-BAND SPECTRAL TRANSPORT	RFWave	$\frac{\text{d}z_t}{\text{d}t} = v_{\theta}(z_t, t \mid m)$		
05	Autoregressive linear prediction LPC PREDICTOR AND NEURAL EXCITATION	LPCNet	$\hat{x}_t = -\sum_{k=1}^P a_{k,t} \hat{x}_{t-k} + \hat{e}_t$		
NOTATION	m : mel spectrogram $\hat{x}$: synthesized waveform	θ : learned parameters G_{θ}, S_{θ} : waveform and spectral heads	$\hat{A}, \hat{\phi}$: STFT magnitude and phase $j^2 = -1$: imaginary unit	h : frame representation t : flow time or sample index z_t, v_{θ} : spectral state and velocity	P, k : LPC order and tap index $a_{k,t}$: linear prediction coefficient $\hat{e}_t$: neural excitation

Experimental Protocol

Budget-controlled training with one retained parameter state per configuration.

EXPERIMENTAL INDEX SETS

$\mathcal{M}$

12 model configurations

$\mathcal{S} = \{42, 43, 44\}$

Evaluation-seed set

$|\mathcal{D}_{\text{test}}| = 525$

Fixed test utterances

$|\mathcal{M} \times \mathcal{S}| = 12 \cdot 3 = 36$

Model-seed evaluation pairs

01

Train and retain

Random initialization with architecture-specific recipes and compute budgets.

$$\theta_m^\star = \text{Select}[\text{Train}(\theta_{m,0}; \mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}, s_{\text{tr}} = 1234, \mathcal{B}_m)], \quad m \in \mathcal{M}$$

$\theta_{\mathbf{m},0}$

Random initialization

$\mathcal{B}_{\mathbf{m}}$

Registered recipe and compute budget

$\theta_{\mathbf{m}}^\star$

One retained parameter state

02

Repeated evaluation

The retained state is applied to every utterance in the fixed test partition.

$$\mathcal{Y}_{m,s} = \left\{ f_{\theta_m^\star}(c_m(x); s) \mid x \in \mathcal{D}_{\text{test}} \right\}, \quad (m, s) \in \mathcal{M} \times \mathcal{S}$$

$\mathcal{Y}_{\mathbf{m},s}$

Generated test outputs

$\mathbf{c}_{\mathbf{m}}(\mathbf{x})$

Model-specific conditioning for utterance x

MEASURED RESPONSE

Audio quality

Perceptual, intelligibility, and signal metrics

Inference speed

Real-time factor under the fixed timing protocol

Model complexity

Parameters, serialized size, and MACs

Quality, Latency and Footprint

Quality over the fixed test split; batch-one inference on NVIDIA B200.

Model	Parameters (M) ↓	Checkpoint (MiB) ↓	PESQ ↑	STOI ↑	B200 RTF ↓
APNet2	31.43	119.88	2.5410	0.9300	0.0095
BigVGAN	14.03	53.51	3.2570	0.9737	0.1573
FreeV	18.22	69.66	3.3394	0.9693	0.0077
HiFi-GAN V1	13.94	53.16	3.3170	0.9748	0.1197
HiFi-GAN V2	0.93	3.54	2.4561	0.9453	0.1140
HiFi-GAN V3	1.46	5.59	2.5765	0.9520	0.0580
LPCNet	1.23	4.70	1.0770	0.5275	11.3633
MeIGAN	4.27	16.27	1.0598	0.4183	0.0742
RNDVoC	3.57	14.94	3.8568	0.9871	0.0559
Vocos	13.53	51.62	3.5653	0.9793	0.0047
VocosFormer	13.78	52.57	3.5125	0.9758	0.0122
RFWave	18.14	69.21	3.7846	0.9809	0.0148

Quality values are means over seeds 42, 43, and 44, with 525 utterances per execution. Bold entries mark the best observed value in each column.

No Single Architecture Dominates

The selected model changes with latency and parameter budgets.

CONSTRAINED SELECTION

Optimize PESQ only over models satisfying both deployment constraints.

$$\mathcal{F}(\tau, B) = \{m \in \mathcal{M} \mid R_{\text{B200}}(m) \leq \tau, P(m) \leq B\},$$
$$m^*(\tau, B) \in \arg \max_{m \in \mathcal{F}(\tau, B)} Q_{\text{PESQ}}(m)$$

Q_{PESQ}(m)
PESQ for model *m*

R_{B200}(m)
NVIDIA B200 real-time factor

P(m)
Parameter count

τ, B
Upper bounds on latency and parameters

OBSERVED LEADERS

OBJECTIVE	MODEL	MEASURED ANCHOR
Highest quality	RNDVoC	PESQ 3.8568 STOI 0.9871
Lowest B200 RTF	Vocos	0.0047
Smallest model	HiFi-GAN V2	0.93 M parameters
Speed quality frontier	RFWave	PESQ 3.7846 RTF 0.0148

EMPIRICAL CONSEQUENCE

Fidelity, latency and footprint select different architectures.

Intervention Protocol

Frozen Study 1 checkpoints; unsupported cells terminate before seed expansion.

Same-lane baselines · compilation · dynamic and weight-only INT8 · INT4 probe · FP16 weights · ONNX
FP32/static INT8 · pruning at 30/50/70% with recovery and matched controls · ODE steps 8/4/2

EXPERIMENTAL INDEX SETS

$|\mathcal{M}| = 12$
Frozen Study 1 checkpoints

$|\mathcal{V}| = 19$
Registered intervention variants

$\mathcal{S} = \{42, 43, 44\}$
Evaluation-seed set

$|\mathcal{A}_{\text{adm}} \times \mathcal{S}| = 100 \cdot 3 = 300$
Boundary-admitted evaluation capsules

01 Gate applicability

Resolve transformed coverage and executable backend before seed expansion.

$$\mathcal{A} = \{(m, v, h) \in \mathcal{M} \times \mathcal{V} \times \mathcal{H} \mid E(m, v, h) = 1, \kappa(m, v, h) > 0\}$$

$\mathbf{E} = \mathbf{1}$
Transformed inference path executes

$\kappa > \mathbf{0}$
Nonzero measured operator coverage

$\mathcal{A}$
Supported model, variant, and lane cells

02 Transform and evaluate

Apply one registered intervention, then execute the complete test partition.

$$\tilde{\theta}_{m,v,h} = T_{v,h}(\theta_m^*), \quad \mathcal{Y}_{m,v,h,s} = \left\{ f_{\tilde{\theta}_{m,v,h}}(c_m(x); s) \mid x \in \mathcal{D}_{\text{test}} \right\}$$

$\theta_{\mathbf{m}}^\star$
Frozen Study 1 parameter state

$\mathbf{T}_{\mathbf{v},\mathbf{h}}$
Variant and execution-context transform

$\mathcal{Y}_{\mathbf{m},\mathbf{v},\mathbf{h},\mathbf{s}}$
Outputs for all 525 test utterances

MEASURED RESPONSE	Quality and signal	Runtime	Complexity and coverage
	Complete Study 1 metric panel	Warm RTF, p50, p95, and transform cost	Parameters, serialized size, MACs, and provider scope

Quality Response to Optimization

Δ PESQ is the optimized score minus its matched baseline. Values near zero indicate preserved quality.

Compilation, FP16 weights, ONNX FP32, and weight-only INT8 stayed within 0.003 PESQ of baseline.

STABLE RESPONSE

Near-baseline quality

Maximum absolute difference from the matched baseline.

Intervention	Maximum $ \Delta$ PESQ
Compilation	$ \Delta_{\text{base}} \leq 0.00040$
FP16 weights	$ \Delta_{\text{base}} \leq 0.00170$
ONNX FP32	$ \Delta_{\text{base}} \leq 0.00013$
INT8 weight-only	$ \Delta_{\text{base}} \leq 0.00250$

These transformations changed PESQ by less than 0.003.

Three-seed means over 525 test utterances. Each delta uses the matched runtime baseline; recovery is measured against the corresponding pruned state.

SENSITIVE RESPONSE

Architecture-dependent quality

Observed PESQ change across supported vocoder configurations.

Intervention	Observed Δ PESQ
Dynamic INT8	$[-0.648, +0.004]$
ONNX static INT8	$[-1.329, -0.004]$
Pruning at 30%	$[-2.740, -0.148]$
RFWave ODE steps 8/4/2	$-0.046 / -0.369 / -1.636$

Recovery after 50% pruning regained up to 1.669 PESQ, but the response was not uniform.

No Single Optimization Dominates

The best intervention changes with model, backend, and permitted quality loss.

CONSTRAINED SELECTION

Select the fastest executed variant subject to explicit quality and deployable-size limits.

$$\mathcal{F}_{m,h}(\varepsilon, \rho) = \{v \in \mathcal{V}_{m,h} \mid \Delta Q_{\text{PESQ}}(m, v) \geq -\varepsilon, D(m, v) \leq \rho D(m, b_m)\},$$
$$v_{m,h}^*(\varepsilon, \rho) \in \arg \min_{v \in \mathcal{F}_{m,h}(\varepsilon, \rho)} R_h(m, v)$$

V_{m,h}
Executed interventions for model *m* on backend *h*

ΔQ_{PESQ}(m,v)
PESQ change from the matched baseline

R_h(m,v)
Real-time factor measured on backend *h*

D(m,v), b_m, ε, ρ
Deployable size, matched baseline, and permitted quality and size limits

OBSERVED CHOICES

OBJECTIVE	CONFIGURATION	MEASURED ANCHOR
Quality-preserving B200 gain	RFWave FP16 weights	1.95× faster 0.50× size ΔPESQ +0.00004
Fastest CPU export	HiFi-GAN V2 ONNX FP32	2.07× faster ΔPESQ −0.00013
Smallest near-baseline artifact	MeiGAN ONNX static INT8	0.27× size 1.47× slower ΔPESQ −0.00438
Maximum raw speed	RFWave 2 ODE steps	4.51× faster ΔPESQ −1.636

EMPIRICAL CONSEQUENCE

Model, backend and constraints jointly determine the best intervention.

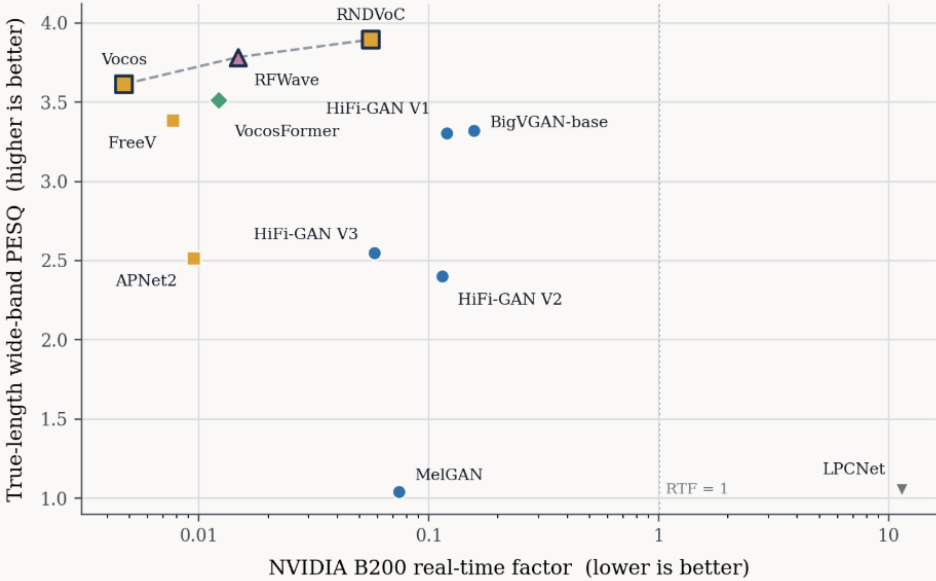
STUDY 3 · COMPARATIVE SYNTHESIS · 1/1

Measured Quality Frontiers

Twelve project-trained models under one dataset, evaluator, and timing protocol.

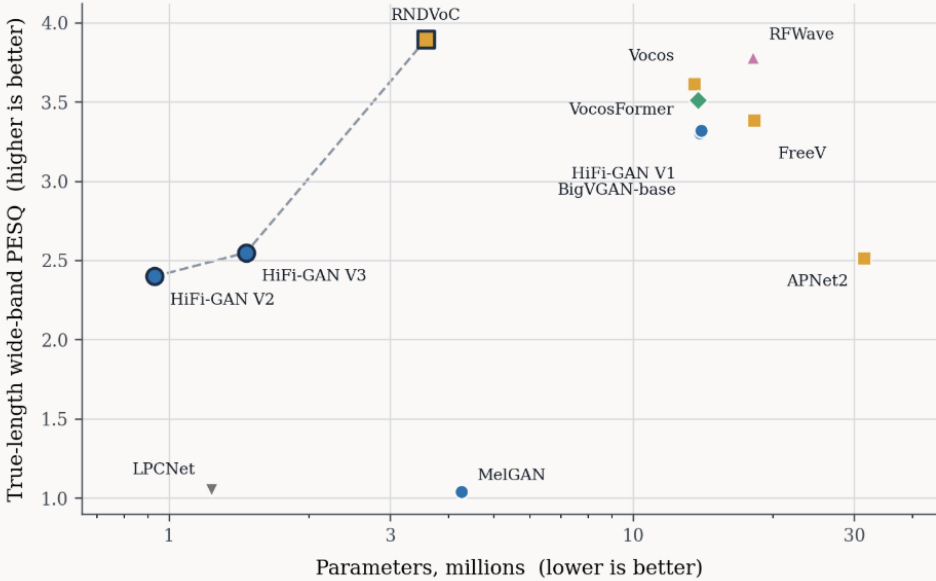
A · INFERENCE SPEED

B200 real-time factor



B · MODEL COMPLEXITY

Trainable parameters



● Time-domain GAN ■ Frame-level Fourier ▲ Flow matching ◆ Hybrid attention ▼ Autoregressive - - - Nondominated frontier

LATENCY-FIRST	FRONTIER MIDPOINT	SHARED FRONTIER ENDPOINT
Vocos RTF 0.0047	RFWave PESQ 3.7845	RNDVoC PESQ 3.8953 · 3.57M parameters

CONCLUSION

Three Studies, One Selection Rule

No architecture or optimization dominates quality, latency, and footprint simultaneously.

STUDY	ANSWER	MEASURED ANCHOR
01 Study 1 Controlled baselines	Different objectives select different architectures.	RNDVoC highest quality · Vocos RTF 0.0047 · HiFi-GAN V2 0.93M parameters
02 Study 2 Controlled interventions	Optimization gain depends on architecture and backend.	RFWave FP16 · 1.95× faster · 0.50× size · ΔPESQ +0.00004
03 Study 3 Comparative synthesis	The nondominated set changes with deployment constraint.	RNDVoC · PESQ 3.8953 · 3.57M parameters · present on both measured frontiers

PROJECT CONCLUSION

Select the **model and intervention jointly** for the target hardware, quality floor, latency ceiling, and footprint budget.

OUTLOOK

Future Work

Three experiments extend the benchmark toward robust, deployment-ready model selection.

	RESEARCH PROGRAM	NEXT EXPERIMENT	DECISION OUTPUT
01	Broader model and data frontier	Add streaming, flow-matching, and neural-codec decoders; evaluate multi-speaker and multilingual corpora.	Measure: architecture ranking across domains.
02	Replicated and perceptual evidence	Train independent replicas of Pareto finalists; conduct randomized, blinded listening tests.	Estimate: training variance and listener preference.
03	Deployment-native optimization	Test QAT, executable INT4, FP8, or FP4 paths, and 2:4 sparse kernels on target runtimes.	Profile: quality, latency, memory, and energy.

RESEARCH OBJECTIVE

A hardware-aware Pareto frontier that remains stable across data, training runs, and runtimes.

FINAL PRESENTATION

Thank You

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